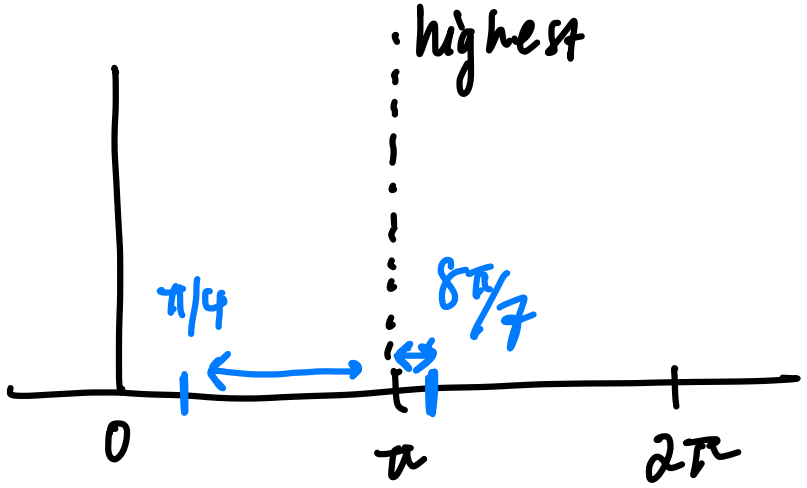
ELEC 221 Lecture 10
DT filters; Introducing the Fourier transform

Tuesday 13 October 2026

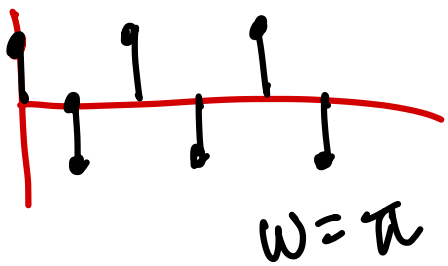
Announcements

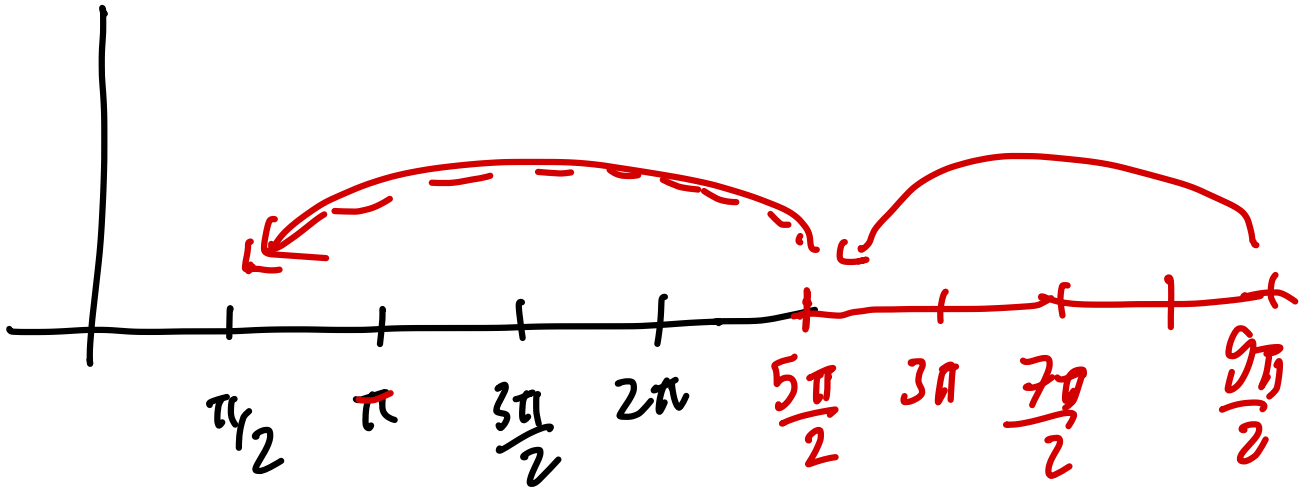


$\omega_1 = \frac{8\pi}{7}$

$\omega_2 = \frac{\pi}{4}$

- Quiz 5 today
- Midterm in 3 weeks. Practice tests will be posted on PrairieLearn by end of week.





Last time

DT Fourier synthesis equation

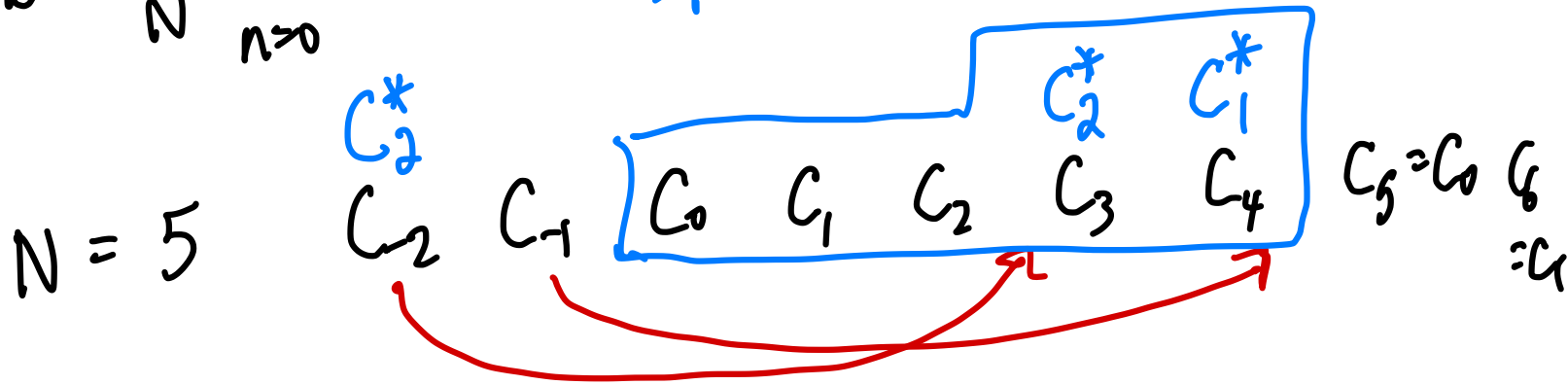
$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk\frac{2\pi}{N}n}$$

DT Fourier analysis equation

$$c_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-jk\frac{2\pi}{N}n}$$

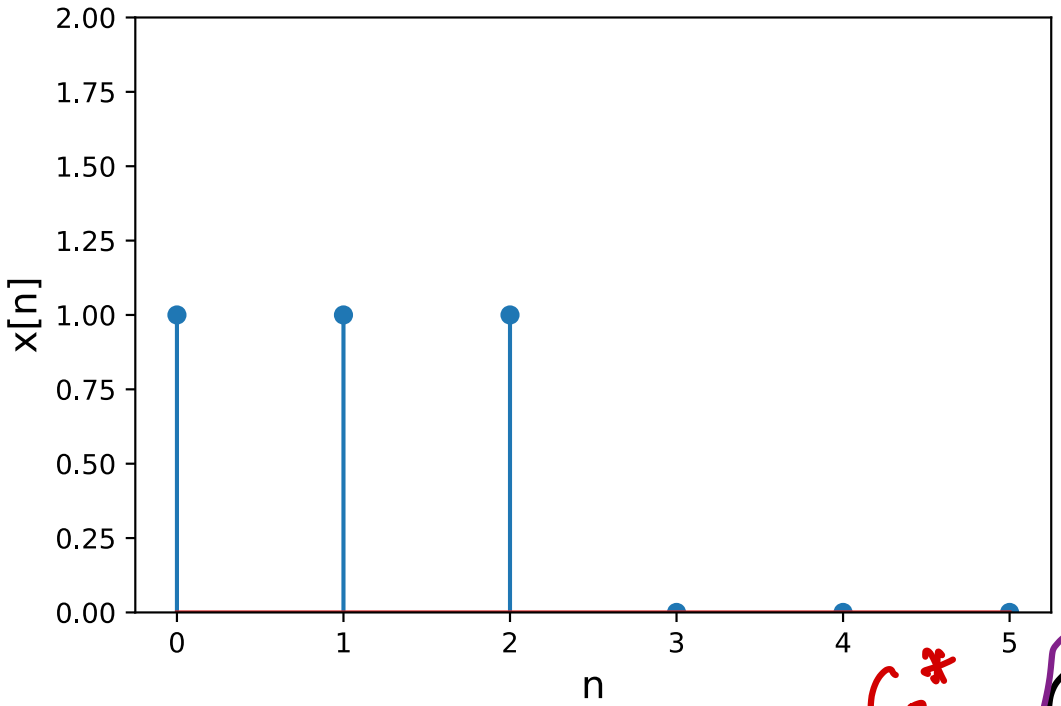
If $x[n]$ real: $c_{-k} = c_k^*$

$N = 5$



Last time

We computed Fourier coefficients of a DT square wave



$N = 6$

$C_{-2}\ C_{-1}\ \boxed{C_0\ \ C_1\ \ C_2\ \ C_3}\ \ C_4\ \ C_5$

C_2^*

$C_{-1} = C_1^*$

$C_0 = \frac{1}{2}$ $C_1 = \frac{1}{6}(1 - \sqrt{3}j)$ $C_2 = 0$ $C_3 = \frac{1}{6}$

Today

Learning outcomes:

- Apply properties of DT Fourier series to determine coefficients of transformed signals
- Distinguish between finite impulse response and infinite impulse response filters in DT
- Distinguish between the CT Fourier series and Fourier transform
- Compute the Fourier spectrum of a CT signal

Properties of DT Fourier coefficients

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

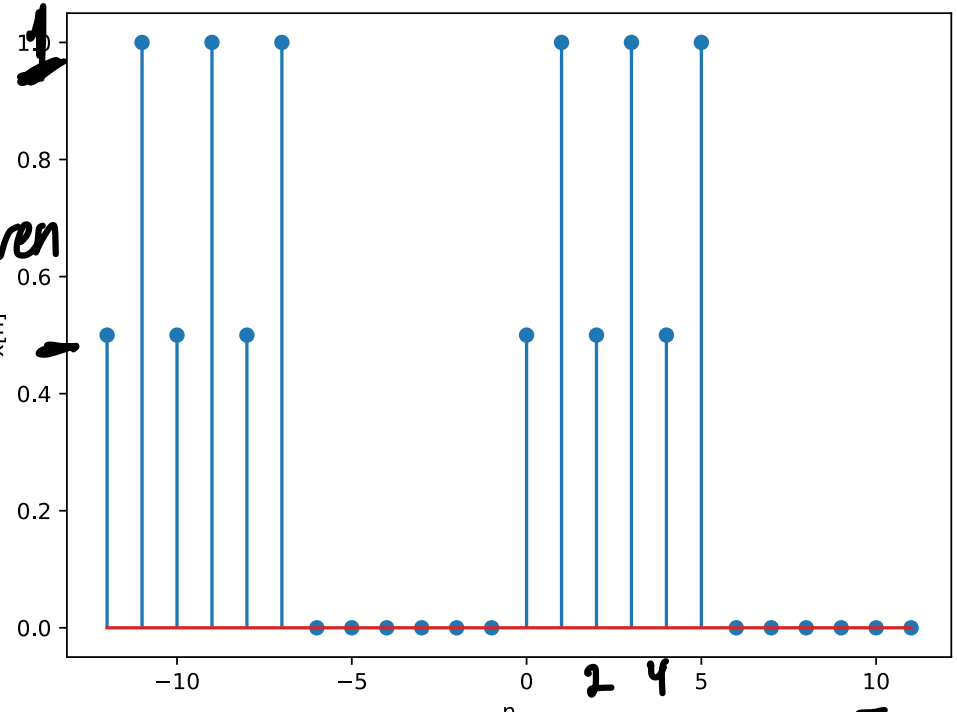
Property	Periodic Signal	Fourier Series Coefficients
	$x[n]$, $y[n]$: Periodic with period N and fundamental frequency $\omega_0 = 2\pi/N$	a_k, b_k: Periodic with period N
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n}x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a^*_{-k}
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m}a_k$ (viewed as periodic with period mN)
Periodic Convolution	$\sum_{r=\langle N \rangle} x[r]y[n-r]$	$Na_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N \rangle} a_l b_{k-l}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running Sum	$\sum_{k=-\infty}^{n} x[k]$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{(1 - e^{-jk(2\pi/N)})}\right)a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$a_k = a^*_{-k}$; $\mathcal{R}e\{a_k\} = \mathcal{R}e\{a_{-k}\}$; $\mathcal{I}m\{a_k\} = -\mathcal{I}m\{a_{-k}\}$; $\lvert a_k\rvert = \lvert a_{-k}\rvert$; $\measuredangle a_k = -\measuredangle a_{-k}$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$x_e[n] = \mathcal{E}v\{x[n]\}$ [x[n] real]; $x_o[n] = \mathcal{O}d\{x[n]\}$ [x[n] real]	$\mathcal{R}e\{a_k\}$; $j\mathcal{I}m\{a_k\}$

Exercise: DT Fourier coefficients

How would we compute the Fourier coefficients for this signal?

Old square wave: $x[n]$

$$x_2[n] = \begin{cases} x[n/2] & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$



$N = 12$

New signal $y[n] = 0.5\,x_2[n] + x_2[n-1]$

find coeffs of this

Exercise: the DT square wave

original: c_0 c_1 c_2 c_3 c_4 c_5 c_6 c_7 ... $\sum_{k=\langle 6 \rangle}$

c_1: $\frac{2\pi n}{6}$, c_2: $2\cdot\frac{2\pi n}{6}$, c_6: same as c_0

new: $x_2[n]$ $\frac{1}{2}c_0$ $\frac{1}{2}c_1$ $\frac{1}{2}c_2$ ... $\frac{1}{2}c_{11}$ $\sum_{k=\langle 12 \rangle}$

$\frac{2\pi n}{12}$, $2\cdot\frac{2\pi n}{12}$

$$0.5\,x_2[n] + x_2[n-1]$$

$$c'_k = \frac{1}{2}c_{k \bmod 6} \qquad c_8 = \frac{1}{2}c_2$$

time scaling ($x_2[n]$) time-shift

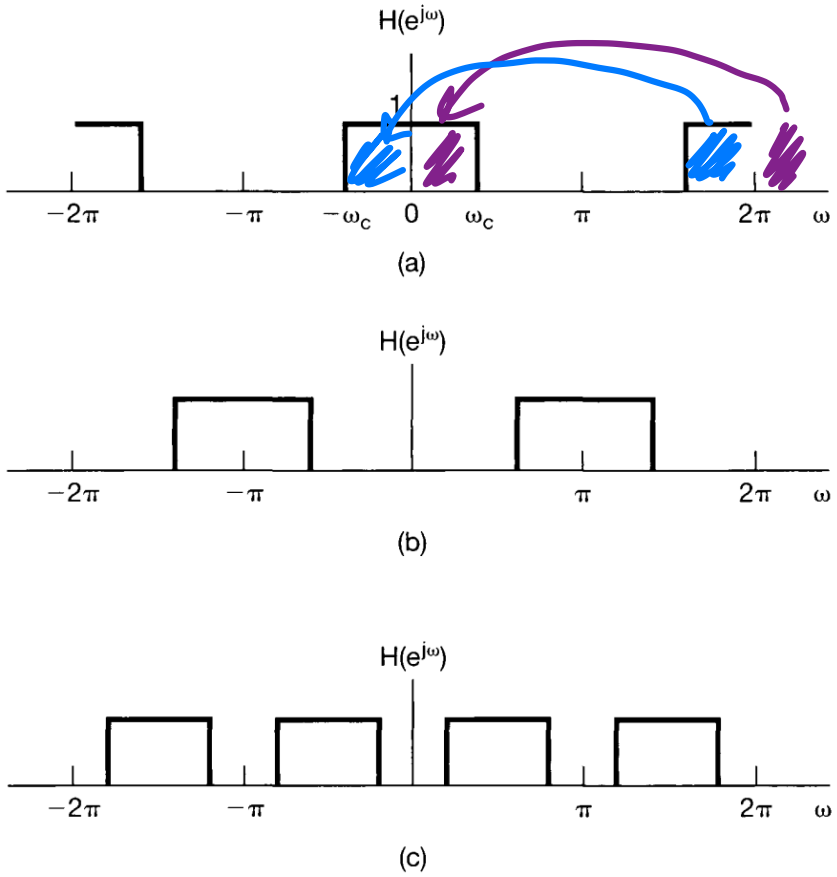
$\Rightarrow$ for combined signal, $\frac{1}{2}c'_k + e^{-jk\frac{2\pi n}{12}}c'_k$

$\frac{1}{2}$: scaling

DT filters

Recall that in DT, the frequency increases up until $\omega = \pi$, then decreases as it approaches 2π.

periodic! (and continuous)



idealized low pass.

high pass.

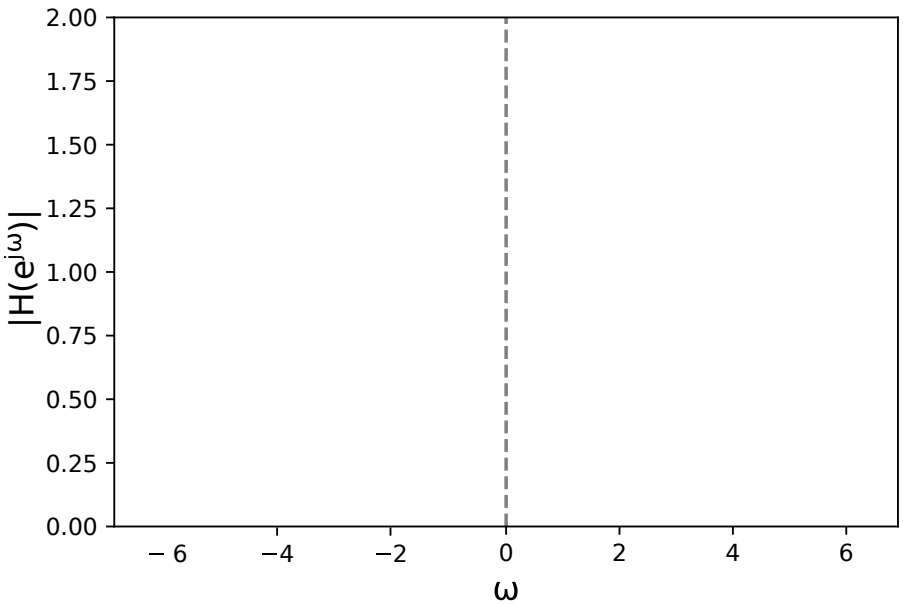
band pass.

Image credit: Oppenheim chapter 3.9.

DT filters

Example: consider the DT filter with frequency response

Sketch below:



DT filters

There are two major categories of DT filters:

1. Infinite impulse response (IIR)
2. Finite impulse response (FIR)

DT filters (IIR)

Example: we can generate a low- or high-pass filter using a system described by a (recursive) first-order difference equation.

$$y[n] - a\,y[n-1] = x[n]$$

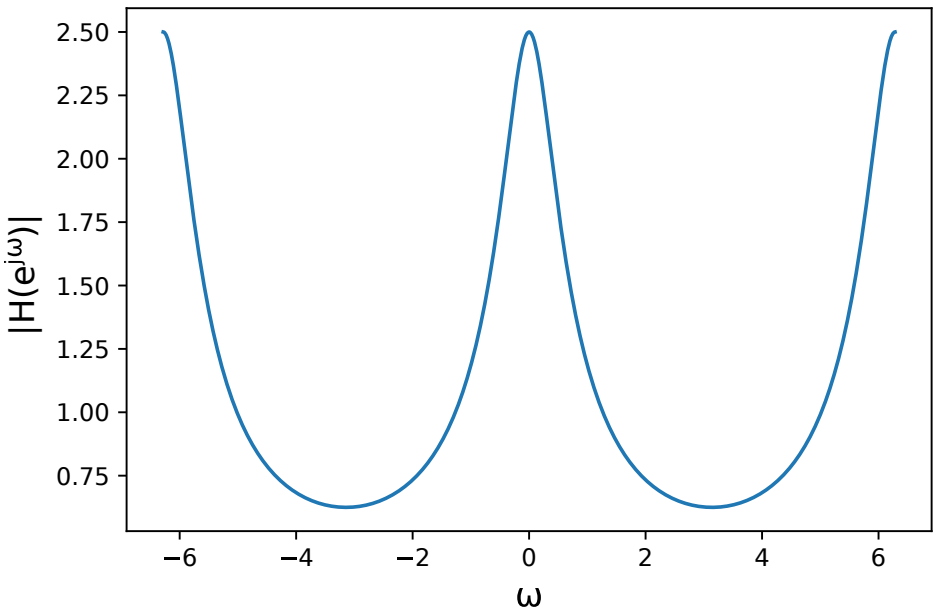
$x[n] = e^{j\omega n}$ input $\qquad y[n] = H(e^{j\omega}) \cdot x[n]$

$$H(e^{j\omega})\,x[n] - a \cdot H(e^{j\omega})\,e^{j\omega(n-1)} = e^{j\omega n}$$

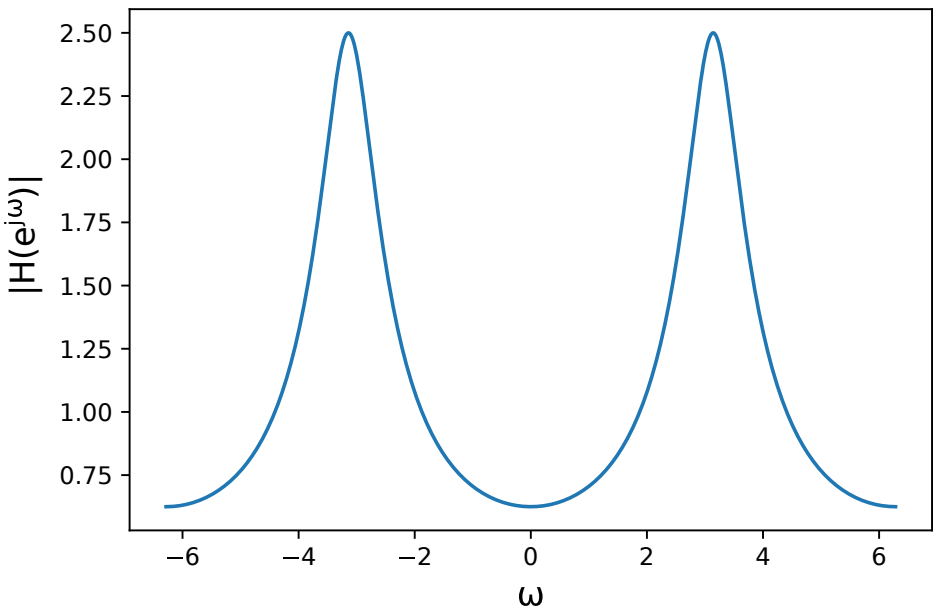
$$H(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}}$$

DT filters (IIR)

Case: $a > 0$



Case: $a < 0$



DT filters (IIR)

To determine the impulse response, assume the system starts at rest ($y[-1] = 0$) and solve recursively:

$$y[n] - a\,y[n-1] = x[n]$$

use $x[0] = \delta[0] \Rightarrow y[0] = 1$

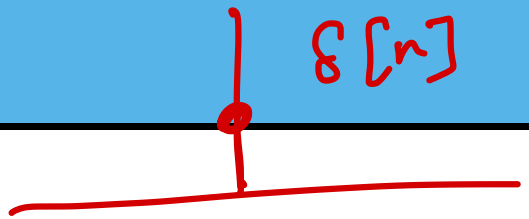
then calculate $y[1]$, $y[2]$ --

$$y[n] = a^n$$

This system has infinite impulse repsonse:

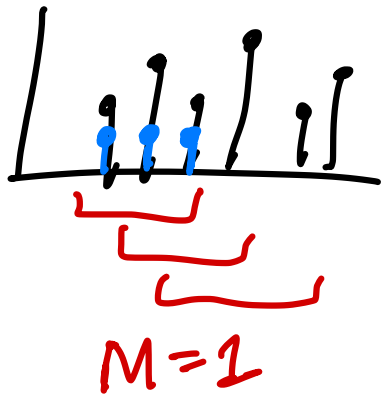
$$h[n] = \underbrace{a^n u[n]}$$ not stable.

DT filters (FIR)



An example of a DT filter with finite impulse response is the moving average with window of size M:

$$y[n] = \frac{1}{2M+1} \sum_{k=-M}^{M} x[n-k]$$



Impulse response: set $x[n] = \delta[n]$

$$h[n] = \frac{1}{2M+1} \sum_{k=-M}^{M} \delta[n-k]$$

This is clearly finite (as a result such filters are stable).

DT filters (FIR)



Let's compute the frequency response:

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} e^{-jk\omega} h[k]$$

Set $M = 1$ (3-point moving average)

$$h[n] = \frac{1}{3}\left(\delta[n-1] + \delta[n] + \delta[n+1]\right)$$

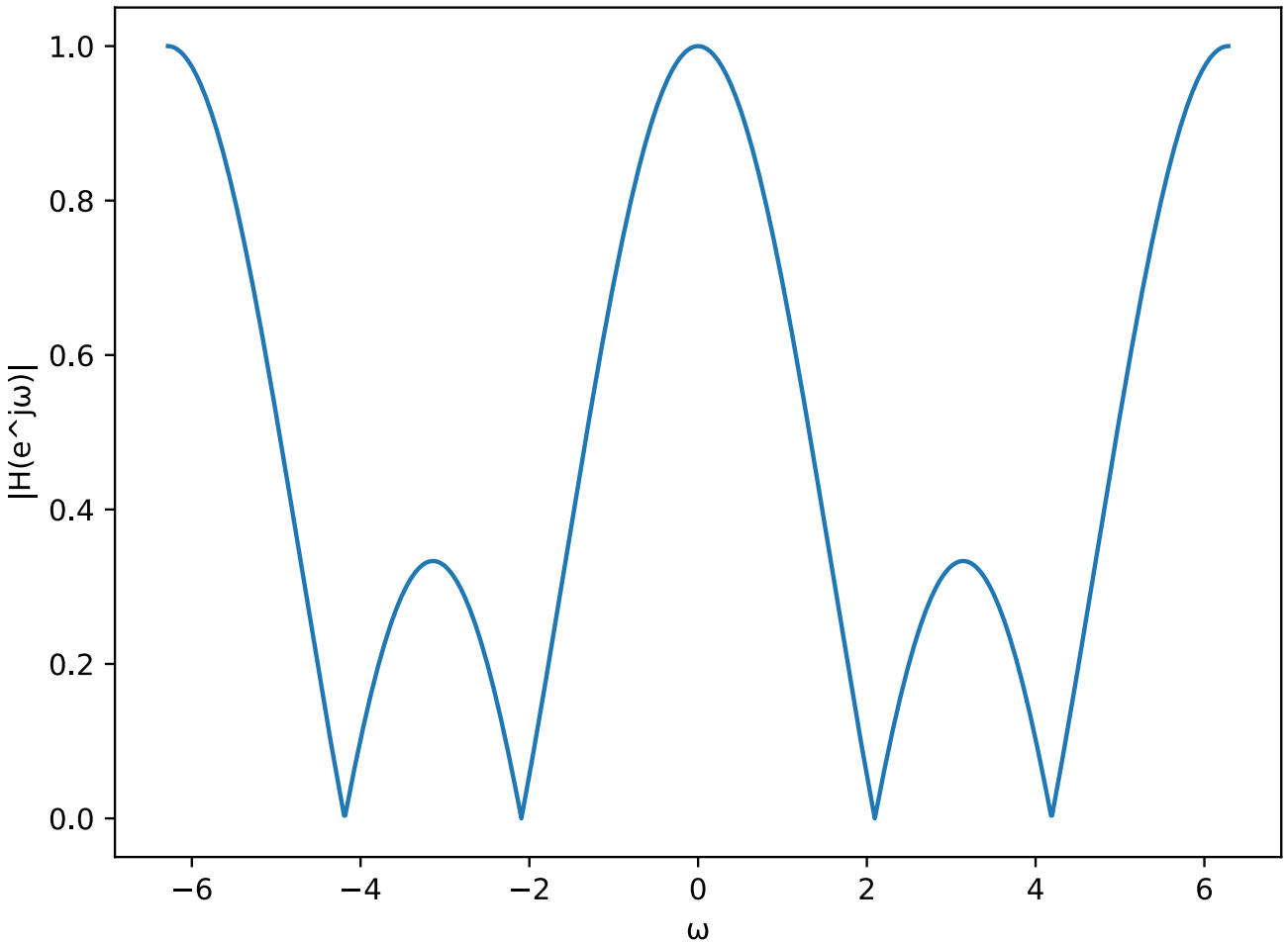
$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} e^{-jk\omega} h[k]$$

$$= \frac{1}{3}\left(e^{-j\omega} + 1 + e^{j\omega}\right)$$

$$= \frac{1}{3}(1 + 2\cos\omega)$$

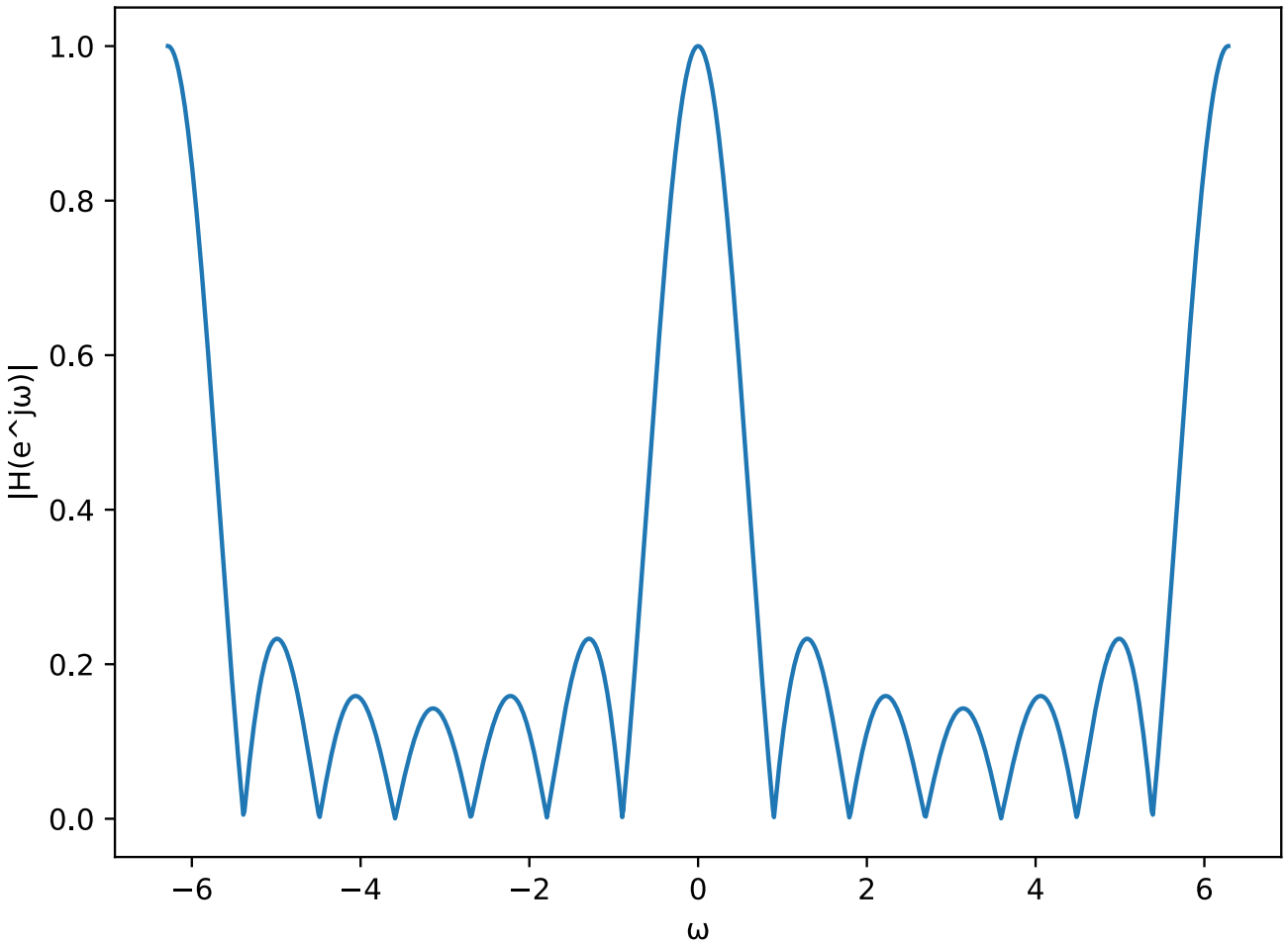
DT filters (FIR)

Example when $M = 1$



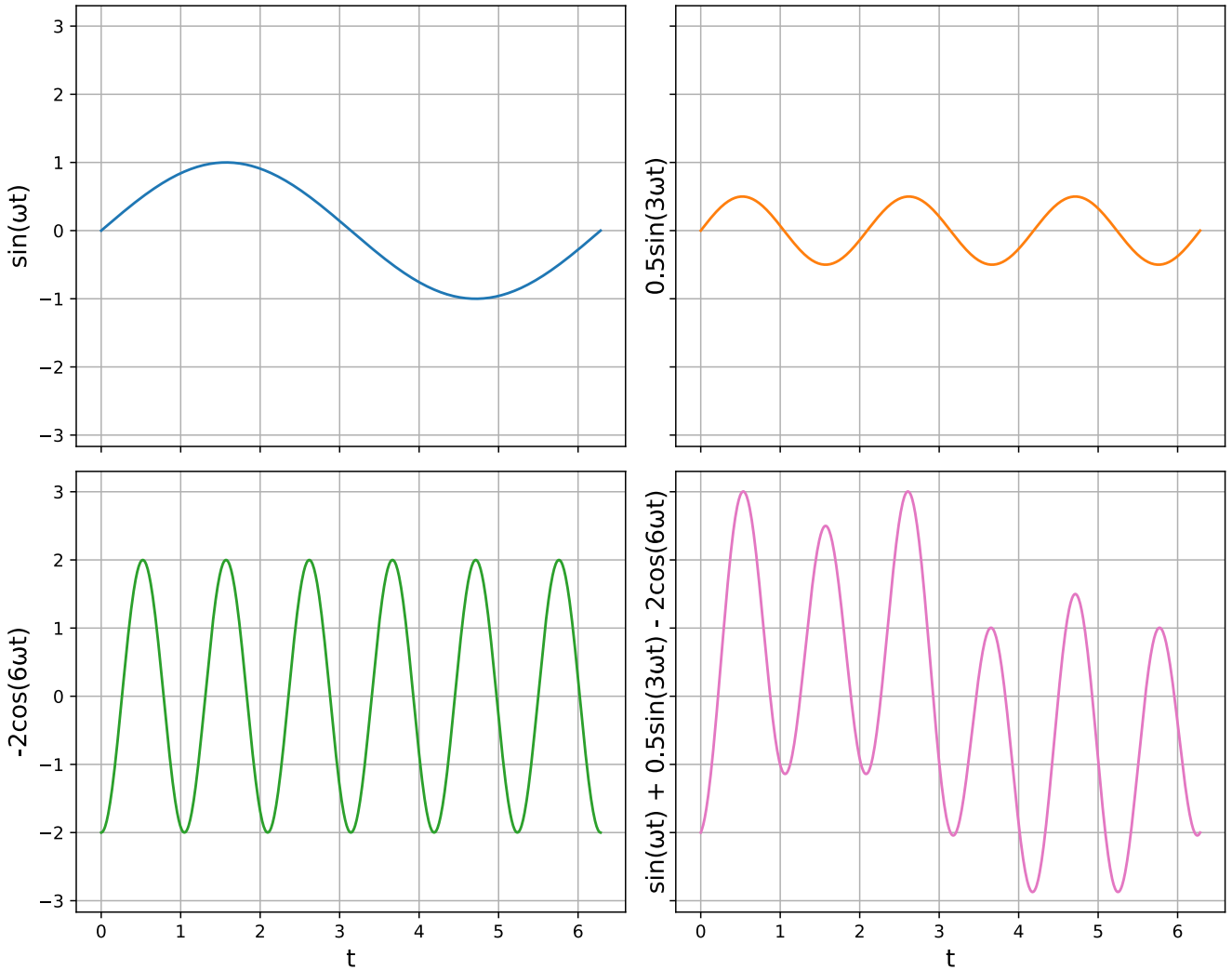
DT filters (FIR)

Example when $M = 3$



Fourier series

We've seen the Fourier series representation of **periodic** signals:



Fourier series

CT synthesis equation:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

CT analysis equation:

$$c_k = \frac{1}{T} \int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

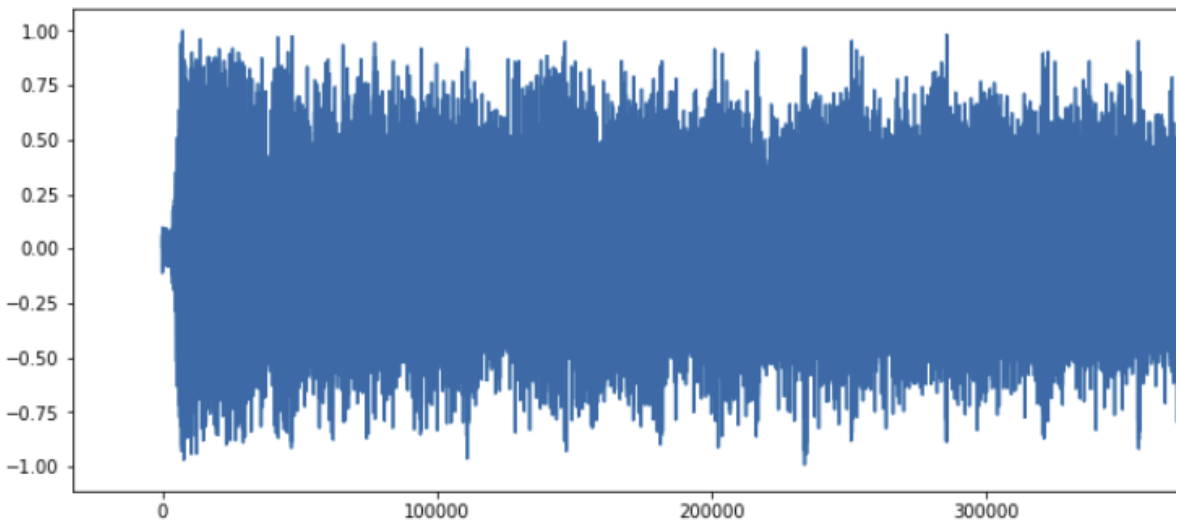
A periodic signal is composed of complex exponential signals with *integer multiples of the fundamental frequency* only (harmonics).

Fourier series

In tutorial assignment 2, you worked with signals that *weren't periodic*:

```
plt.plot(audio)
```

```
[<matplotlib.lines.Line2D at 0x7f940880e370>]
```



But, we were still doing *something* with Fourier analysis:

```
fourier_spectrum = np.fft.rfft(audio)
```

The Fourier transform

The **Fourier transform** generalizes the Fourier series to **aperiodic signals**. It involves all possible frequencies.

Fourier series:

Fourier transform:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \qquad X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

How do we get here?

For next time

Content:

→ CT Fourier transform.

- Fourier transform for periodic signals
- Properties of the CT Fourier *transform*
- Time/frequency duality

Action items:

1. Work on practice problems

Recommended reading:

- From today's class: Oppenheim 3.7, 3.11, 4.0-4.2
- Suggested problems: 4.1, 4.2a, 4.21abei, 4.22abde
- For next class: Oppenheim 4.2-4.4